

$$\begin{array}{ccc}
 \mathbb{D}^* & \xrightarrow{z \mapsto \lambda_0 z} & \mathbb{D}^* \\
 \exp \uparrow & & \\
 \mathbb{H}_\ell & \xrightarrow{\zeta \mapsto \zeta + v_0} & \mathbb{H}_\ell \\
 L_v \downarrow & & \downarrow L_v \\
 \mathbb{H}_\ell & \xrightarrow{\zeta \mapsto \zeta + v} & \mathbb{H}_\ell
 \end{array}$$